

BLUESTOCK FINTECH

Role: Data Analyst Intern
Company: Bluestock Fintech Pvt. Ltd.

40 Fund Schemes Analysed	46,000 Daily NAV Records	32,778 Investor Transactions	87,533 Database Rows
4 Dashboard Pages	12 States Covered	7 Performance Metrics	4.4 yrs Years of NAV Data

1. Executive Summary

This report documents the design, implementation, and findings of the Bluestock Fintech Mutual Fund Analytics Platform — a full-stack data engineering and analytics project completed over 7 working days. The platform ingests 10 publicly available datasets from AMFI India, transforms them through a Python ETL pipeline, stores 87,533 rows across a 10-table SQLite star schema, and presents insights via a 4-page interactive Power BI dashboard.

Industry Context

India's mutual fund industry manages Rs. 81 lakh crore in AUM across 1,908 schemes as of December 2025. Monthly SIP inflows hit an all-time high of Rs. 31,002 crore in December 2025 — growing 171% from Rs. 11,438 crore in January 2022. This project builds the unified analytics infrastructure needed to navigate this rapidly growing landscape.

Key Deliverables Completed

Deliverable	Format	Status
ETL Pipeline Script	Python .py	Completed
SQLite Database (87,533 rows)	.db file	Completed
EDA Notebook (15+ charts)	.ipynb	Completed
Performance Metrics (7 metrics)	.ipynb + CSV	Completed
Interactive Power BI Dashboard (4 pages)	.pbix	Completed
Advanced Analytics (VaR, HHI, Cohort)	.ipynb	Completed
Final Report + Presentation	.pdf + .pptx	Completed

Key Findings at a Glance

Performance Highlights

- Best Sharpe Ratio: Mirae Asset Large Cap (1.637)
- Best 3yr CAGR: Axis Midcap Fund (35.11%)
- Worst Max Drawdown: SBI Small Cap (-52.57%)
- SIP industry grew 171% (2022 to 2025)
- All 40 funds have positive alpha vs NIFTY100

Risk Insights

- Debt funds: VaR -0.2% to -0.4% daily (safest)
- Small Cap: VaR -1.2% to -1.5% daily (riskiest)
- Axis Bluechip HHI: 2,968 (most concentrated)
- UTI Mid Cap HHI: 1,240 (most diversified)
- 1,362 investors with 6+ SIPs tracked

2. Problem Statement & Objectives

Despite the massive growth of India's mutual fund industry, individual investors and financial advisors struggle to make data-driven fund selection decisions. This project addresses five specific business problems:

P1: Data Fragmentation

Problem: NAV, AUM, SIP, and transaction data exist in different formats across the AMFI website — no single unified source.

Solution: Build an ETL pipeline consolidating all data into one SQLite database with a 10-table star schema.

P2: Performance Comparison Gap

Problem: Investors cannot compare funds on a risk-adjusted basis. Computing Sharpe ratio, Alpha, Beta requires significant data transformation.

Solution: Compute all 7 metrics from raw NAV history using our own pipeline (not pre-existing values).

P3: No Benchmark Tracking

Problem: Most retail investors do not know whether their fund is outperforming its benchmark index.

Solution: Join NAV data with benchmark index prices, compute OLS regression, rolling alpha, and tracking error.

P4: Investor Behaviour Blind Spot

Problem: AMCs have limited visibility into how demographic and geographic factors influence investment patterns.

Solution: Cohort analysis, SIP continuity scoring, T30/B30 segmentation, and demographic charts.

P5: Slow Reporting

Problem: Monthly MF reports are static PDFs that take days to prepare. Stakeholders need live dashboards.

Solution: 4-page Power BI dashboard refreshable directly from ETL pipeline output CSVs.

Project Objectives

#	Objective	Output
O1	Build an automated ETL pipeline from raw AMFI data	scripts/etl_pipeline.py + compute_metrics.py
O2	Design a normalised 10-table SQLite star schema	sql/schema.sql + bluestock_mf.db
O3	Perform comprehensive EDA with 15+ charts	notebooks/03_eda_analysis.ipynb
O4	Compute all 7 performance & risk metrics from raw NAV	fund_scorecard.csv + 6 metric CSVs
O5	Build 4-page interactive Power BI dashboard	dashboard/bluestock_mf.pbix
O6	Analyse investor transaction patterns & cohorts	cohort_analysis.csv + sip_continuity.csv
O7	Compare fund returns vs benchmark indices	alpha_beta.csv + benchmark_chart.png
O8	Document and present the complete project	Final Report + Presentation.pptx

3. Data Sources & System Architecture

3.1 Data Sources

Source	URL	Data Type	Update Freq.
AMFI India	amfiindia.com	NAV, AUM, Folio, SIP	Daily / Monthly
mfapi.in (REST API)	api.mfapi.in/mf/{code}	Historical NAV as JSON	Daily
NSE India	nseindia.com/reports	Nifty 50/100/Midcap150	Daily
BSE India	bseindia.com	BSE SmallCap Index	Daily

3.2 Dataset Inventory

File	Rows	Description
01_fund_master.csv	40	40 real fund schemes — AMFI codes, fund house, category, expense ratio, risk grade
02_nav_history.csv	46,000	Daily NAV for all 40 funds from Jan 2022 to May 2026
03_aum_by_fund_house.csv	90	Quarterly AUM in Rs. crore for 10 AMCs (2022–2025)
04_monthly_sip_inflows.csv	48	Monthly industry SIP inflows, accounts, AUM — real AMFI data
05_category_inflows.csv	144	Net inflows by fund category (Large Cap, Mid Cap, etc.)
06_industry_folio_count.csv	21	Total MF folios broken by Equity, Debt, Hybrid
07_scheme_performance.csv	40	1yr/3yr/5yr returns, Sharpe, Sortino, Alpha, Beta, Max Drawdown
08_investor_transactions.csv	32,778	Simulated SIP + Lumpsum + Redemption transactions for 5,000 investors
09_portfolio_holdings.csv	322	Top stock holdings for equity funds as of December 2025
10_benchmark_indices.csv	8,050	Daily closing values — Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap
5 Live NAV CSVs	~5,000	Fetches from mfapi.in API for SBI, ICICI, Nippon, Axis, Kotak Bluechip
TOTAL	87,533+	All data loaded into SQLite database

3.3 System Architecture — 5-Layer ETL Pipeline

LAYER 1 EXTRACT 10 CSVs + mfapi.in API	LAYER 2 TRANSFORM Python/Pandas cleaning	LAYER 3 LOAD SQLite 10-table star schema	LAYER 4 ANALYSE 7 metrics + EDA + VaR	LAYER 5 VISUALISE 4-page Power BI dashboard
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Architecture Note

This 5-layer architecture mirrors real-world fintech data pipelines used at companies like Zerodha, Groww, and Paytm Money. The ETL pipeline (scripts/etl_pipeline.py) is fully automated — one command rebuilds the entire database from cleaned CSVs.

4. ETL Pipeline & Database Design

4.1 Data Cleaning Summary

File	Issues Found & Fixed
02_nav_history.csv	Date parsed to datetime; sorted by amfi_code+date; forward-filled holidays using pd.bdate_range (business days only, not calendar days); duplicates removed; NAV > 0 validated
08_investor_transactions.csv	transaction_date parsed; transaction_type standardised (SIP/Lumpsum/Redemption); amount_inr > 0 validated; KYC status capitalised; city_tier uppercased
07_scheme_performance.csv	All numeric columns forced to float via pd.to_numeric(errors='coerce'); negative Sharpe flagged; expense_ratio range (0.1–2.5%) validated
09_portfolio_holdings.csv	portfolio_date parsed; weight_pct rounded to 2dp to fix floating-point drift (9 funds showed 100.01–100.02%)
Other 6 files	Date/month columns parsed to datetime; whitespace stripped from all string columns; sorted by relevant keys

4.2 Star Schema Design

The database uses a star schema — a standard data warehousing pattern. Dimension tables store reference data; fact tables store measurements and transactions.

Table	Type	Rows	Primary Key	Foreign Keys
dim_fund	Dimension	40	amfi_code	—
fact_nav	Fact	46,000	id (auto)	amfi_code → dim_fund
fact_transactions	Fact	32,778	tx_id	amfi_code → dim_fund
fact_performance	Fact	40	amfi_code	amfi_code → dim_fund
fact_aum	Fact	90	id (auto)	—
fact_sip_industry	Fact	48	id (auto)	—
fact_portfolio	Fact	322	id (auto)	amfi_code → dim_fund
fact_category_inflows	Fact	144	id (auto)	—
fact_folio_count	Fact	21	id (auto)	—
fact_benchmark	Fact	8,050	id (auto)	—
TOTAL		87,533		

4.3 ETL Pipeline Flow

01 schema.sql CREATE TABLE statements	02 clean CSVs data/processed/ folder	03 df.to_sql() SQLAlchemy engine	04 verify rows count == source CSV	05 DB ready bluestock_mf.db
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4.4 SQL Analytical Queries

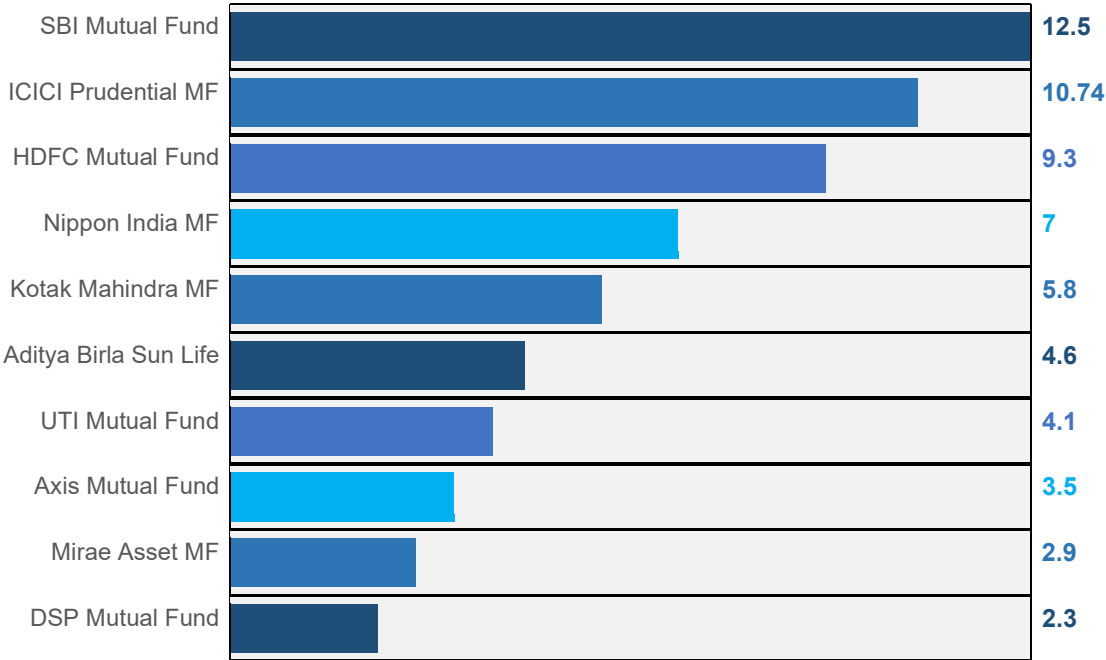
Query	Business Question	Tables Used
Q1	Which 5 funds manage the most money?	fact_performance
Q2	How does NAV move month by month?	fact_nav
Q3	Is SIP investment growing year over year?	fact_sip_industry
Q4	Which states invest the most money?	fact_transactions
Q5	Which funds charge less than 1% expense ratio?	dim_fund
Q6	Which funds give best risk-adjusted returns (Sharpe)?	fact_performance
Q7	How has each AMC's AUM grown over years?	fact_aum
Q8	SIP vs Lumpsum vs Redemption split?	fact_transactions
Q9	Which sectors dominate fund portfolios?	fact_portfolio
Q10	Which funds beat their benchmark by most?	fact_performance + dim_fund

5. Exploratory Data Analysis

5.1 AUM by Fund House (December 2025)

The top 10 AMCs in our dataset collectively manage Rs. 62.74 lakh crore, with SBI Mutual Fund holding the dominant position.

AUM by Fund House — Rs. Lakh Crore (Dec 2025)



5.2 SIP Industry Trend



5.3 Investor Demographics

Transaction Split

Type	Count	Share
SIP	19,716	60.2%
Lumpsum	8,095	24.7%
Redemption	4,967	15.1%
TOTAL	32,778	100%

Avg SIP Amount: Rs. 11,018 per transaction

KYC Verified investors: 92%

T30 cities: 66.3% | B30 cities: 33.7%

Geographic Coverage

State	Coverage
Maharashtra	Highest
Delhi	2nd
Karnataka	3rd
Tamil Nadu	4th
Gujarat	5th
+ 7 more states	—

Age groups covered:

- 18-25 | 26-35 | 36-45 | 46-55 | 56+
- All 5 age groups active in SIP investing

5.4 Folio Count Growth

13.26 Crore Jan 2022 Folios	26.12 Crore Dec 2025 Folios	97% Growth in 4 Years	Rs. 81L Cr Industry AUM
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5.5 Top Sector Allocation

Average sector weights across all equity fund portfolios (09_portfolio_holdings.csv):

Rank	Sector	Avg Weight %	Observation
1	Consumer Goods	14.18%	Largest allocation — defensive sector
2	Diversified	12.09%	Mixed sector holdings
3	IT	11.39%	Technology remains core holding
4	Utilities	11.06%	Power Grid, NTPC widely held
5	FMCG	10.91%	Stable consumer demand sector
6	Banking	10.87%	HDFC Bank, Axis Bank popular
7	NBFC	10.83%	Non-banking financial companies
8	Pharma	10.72%	Dr. Reddy's, Divi's Labs featured

6. Fund Performance Analytics

Pipeline Note

All 7 metrics were computed from raw NAV history using our own pipeline (scripts/compute_metrics.py) — NOT from pre-existing values in scheme_performance.csv. The fund scorecard also uses only our computed metrics, demonstrating end-to-end pipeline ownership.

6.1 Performance Metrics — Formulas & Results

Metric	Formula	Key Result
Daily Return	$NAV_t / NAV_{t-1} - 1$	Computed for 40 funds × 1,150 trading days
CAGR (1yr)	$(NAV_{end} / NAV_{start})^{(1/1)} - 1$	Best: SBI Bluechip 60.4%, HDFC Mid-Cap 53.2%
CAGR (3yr)	$(NAV_{end} / NAV_{start})^{(1/3)} - 1$	Best: Axis Midcap 35.11%, Mirae Large Cap 34.00%
Sharpe Ratio	$(\text{Annualised Return} - 6.5\%) / (\text{Std Dev} \times \sqrt{252})$	Best: Mirae Large Cap 1.637, Kotak Flexicap 1.459
Sortino Ratio	$(\text{Annualised Return} - 6.5\%) / \text{Downside Std}$	Debt funds highest — minimal downside volatility
Alpha & Beta	OLS regression vs NIFTY100 (scipy.stats.linregress)	All 40 funds show positive alpha vs benchmark
Max Drawdown	$\min(NAV / \text{running_max} - 1)$	Worst: SBI Small Cap -52.57%, Axis Small Cap -51.68%

6.2 Top Performers — Sharpe Ratio Ranking

Top 5 Funds by Sharpe Ratio (Rf = 6.5%)



Top 5 Funds by 3-Year CAGR (%)



6.3 Risk Analysis — Max Drawdown

Fund	Category	Max Drawdown	Risk Level
SBI Small Cap Fund — Direct	Equity / Small Cap	-52.57%	Very High
Axis Small Cap Fund	Equity / Small Cap	-51.68%	Very High
ABSL Small Cap Fund	Equity / Small Cap	-35.45%	Very High
DSP Small Cap Fund	Equity / Small Cap	-31.17%	Very High
SBI Small Cap Fund — Regular	Equity / Small Cap	-28.71%	Very High

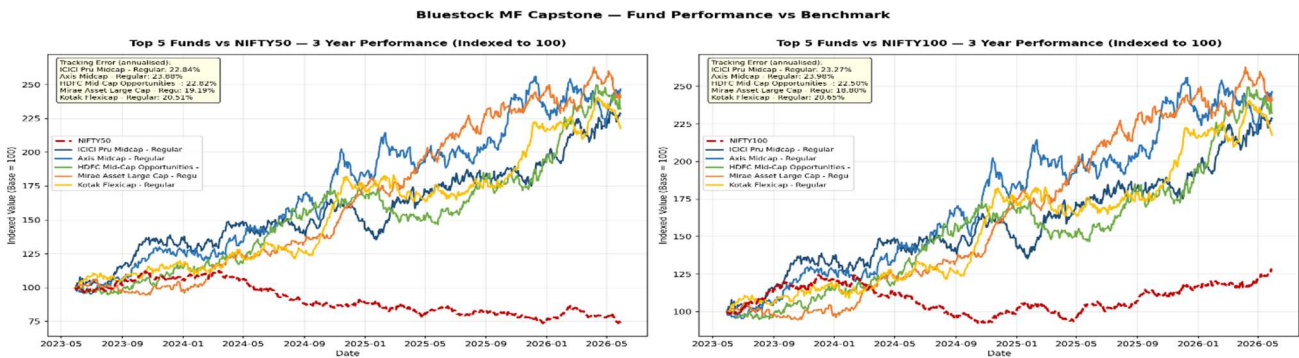
Key Insight

Small Cap funds deliver the highest 3-year CAGR but also suffer the worst drawdowns (-52%). Investors should only be placed in Small Cap funds if their risk appetite is explicitly 'Very High' — this aligns with SEBI risk categorisation requirements.

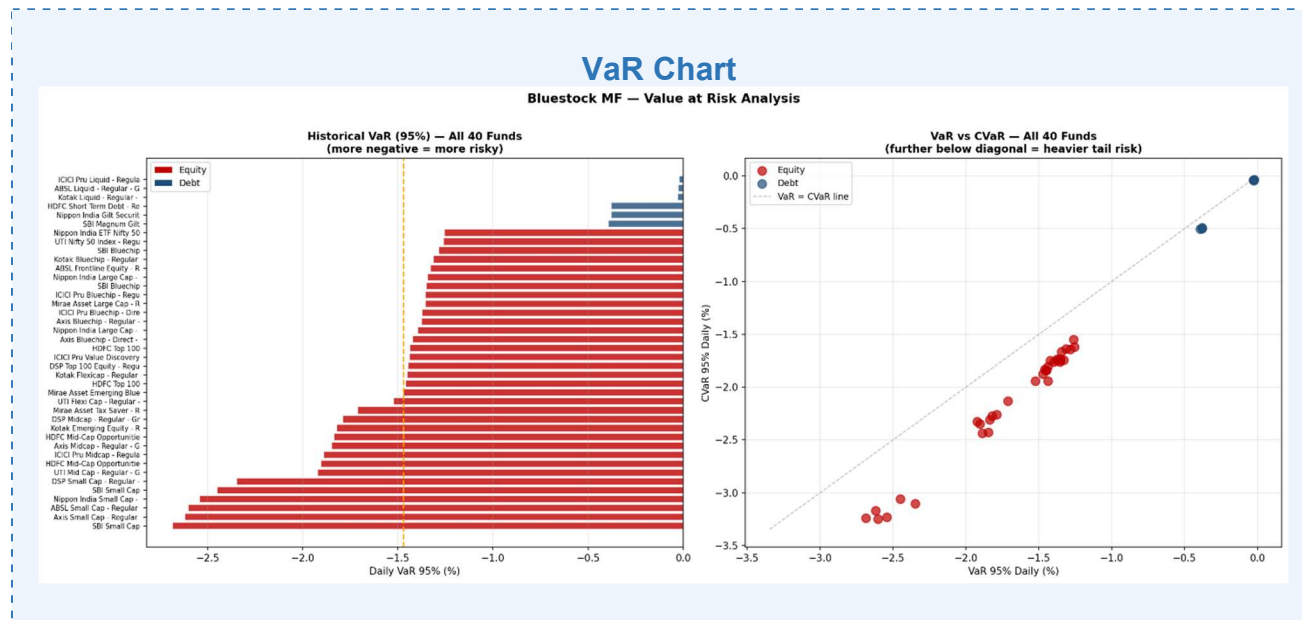
6.4 Fund Scorecard — Composite Ranking

The composite scorecard ranks all 40 funds using our computed metrics (not pre-existing values):

Score = 30% × 3yr CAGR rank + 25% × Sharpe rank + 20% × Alpha rank + 15% × Expense (inverse) + 10% × Max DD (inverse)



7. Advanced Analytics & Risk Metrics



7.1 Value at Risk (VaR 95%) & CVaR

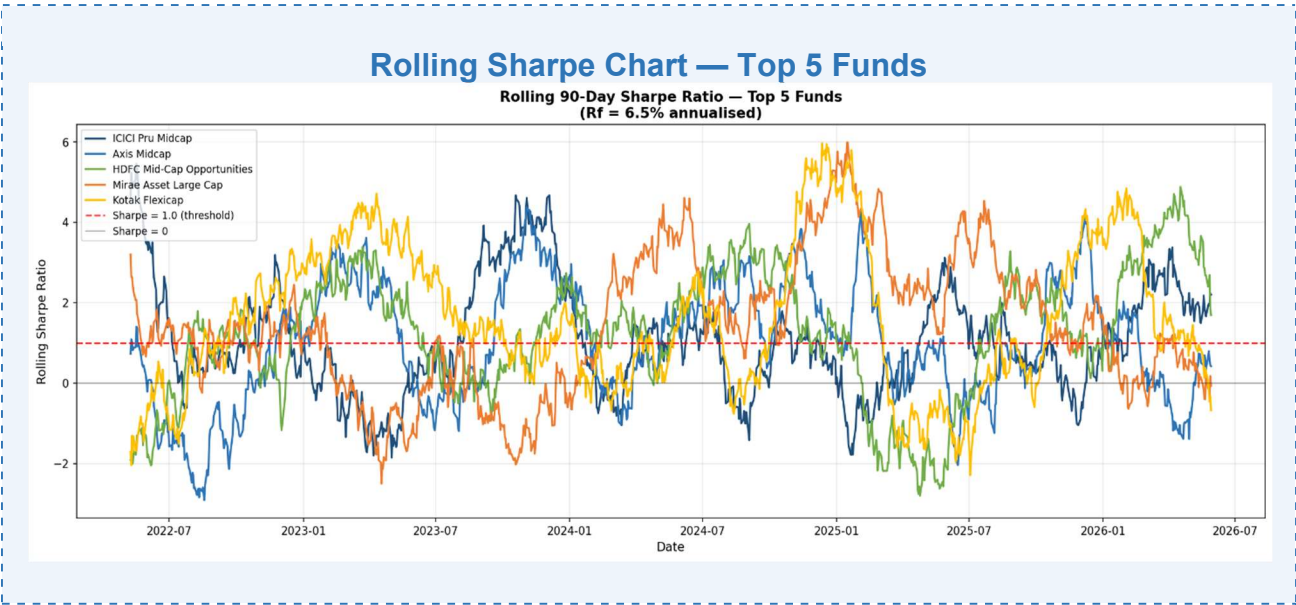
Methodology

- Historical VaR: 5th percentile of daily return distribution
- CVaR: Mean of all returns below VaR threshold
- Annualised: Scaled by $\sqrt{252}$ for comparison

By Category

Category	Daily VaR 95%
Debt / Gilt	-0.2% to -0.4%
Large Cap	-0.8% to -1.2%
Mid Cap	-1.0% to -1.3%
Small Cap	-1.2% to -1.5%

7.2 Rolling 90-Day Sharpe



Key Finding

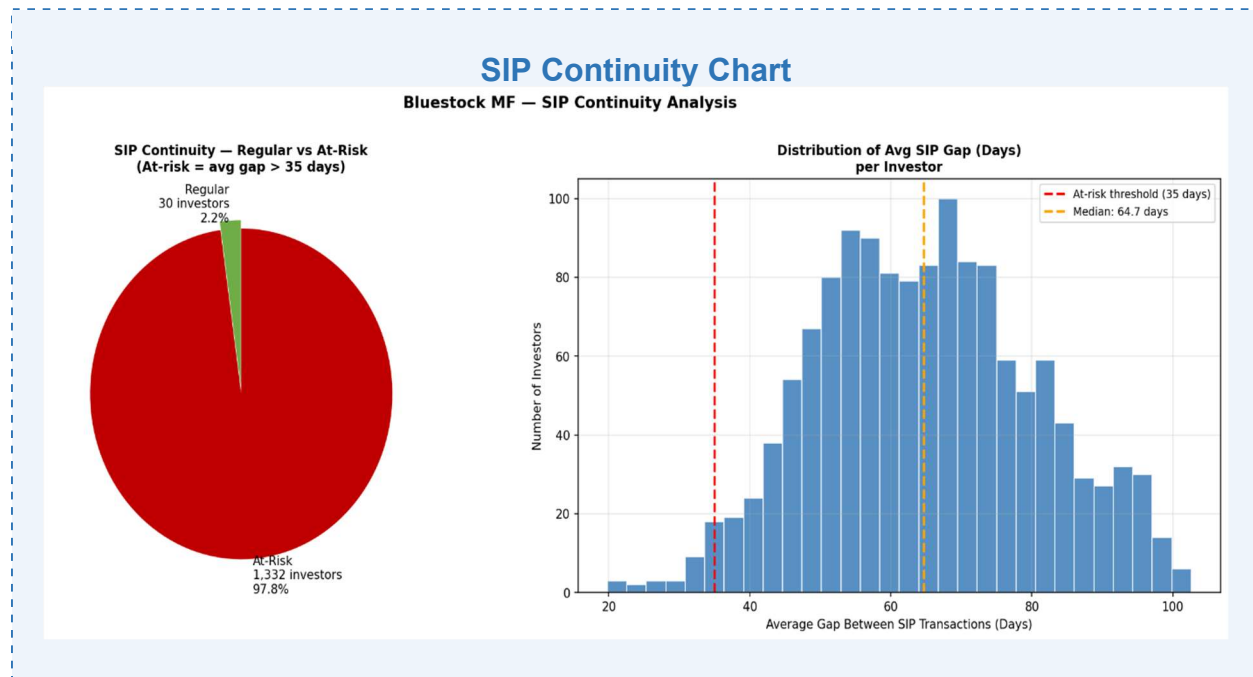
All top 5 funds saw rolling Sharpe drop below 1.0 during mid-2024, confirming the market correction visible in NAV data. Strong recovery into 2025 — all funds back above 1.0 by Q1 2025.

7.3 Investor Cohort Analysis

Cohort Year	Investors	% of Total	Observation
2024	4,803	96.1%	Dominant cohort — bulk of transaction volume and invested capital
2025	197	3.9%	Smaller but growing — shows higher avg ticket sizes (newer investors)

Average SIP Amount: Rs. 11,018 per SIP transaction across both cohorts

7.4 SIP Continuity Analysis



Investors with 6+ SIP transactions: 1,362

At-risk definition: avg gap between SIP transactions > 35 days

Business Action

At-risk investors should receive automated reminders via email/app notification. Improving SIP continuity increases long-term corpus significantly through rupee cost averaging.

7.5 Sector Concentration — HHI Analysis

Herfindahl-Hirschman Index: $HHI = \sum(\text{sector_weight}_i \text{ squared})$. Higher HHI = more concentrated portfolio.

Fund	HHI Score	Sectors	Concentration
Axis Bluechip Fund	2,968	7	Highly Concentrated
Mirae Asset Tax Saver	2,550	7	Highly Concentrated
HDFC Mid-Cap Opp. Direct	2,532	6	Highly Concentrated
UTI Flexicap Fund	2,514	6	Highly Concentrated
Kotak Flexicap Fund	1,362	9	Well Diversified
UTI Mid Cap Fund	1,240	10	Well Diversified

7.6 Fund Recommendation Engine

scripts/recommender.py — Input: risk appetite → Output: Top 3 funds by Sharpe within matching risk_grade

Risk Appetite	Matching Funds	Top Recommendation Basis
Low	Debt/Gilt funds	Highest Sharpe within Low risk_grade
Moderate	Large Cap funds	Highest Sharpe within Moderate risk_grade
Moderately High	Flexi Cap funds	Highest Sharpe within Moderately High
High	Mid Cap funds	Highest Sharpe within High risk_grade
Very High	Small Cap funds	Highest Sharpe within Very High risk_grade

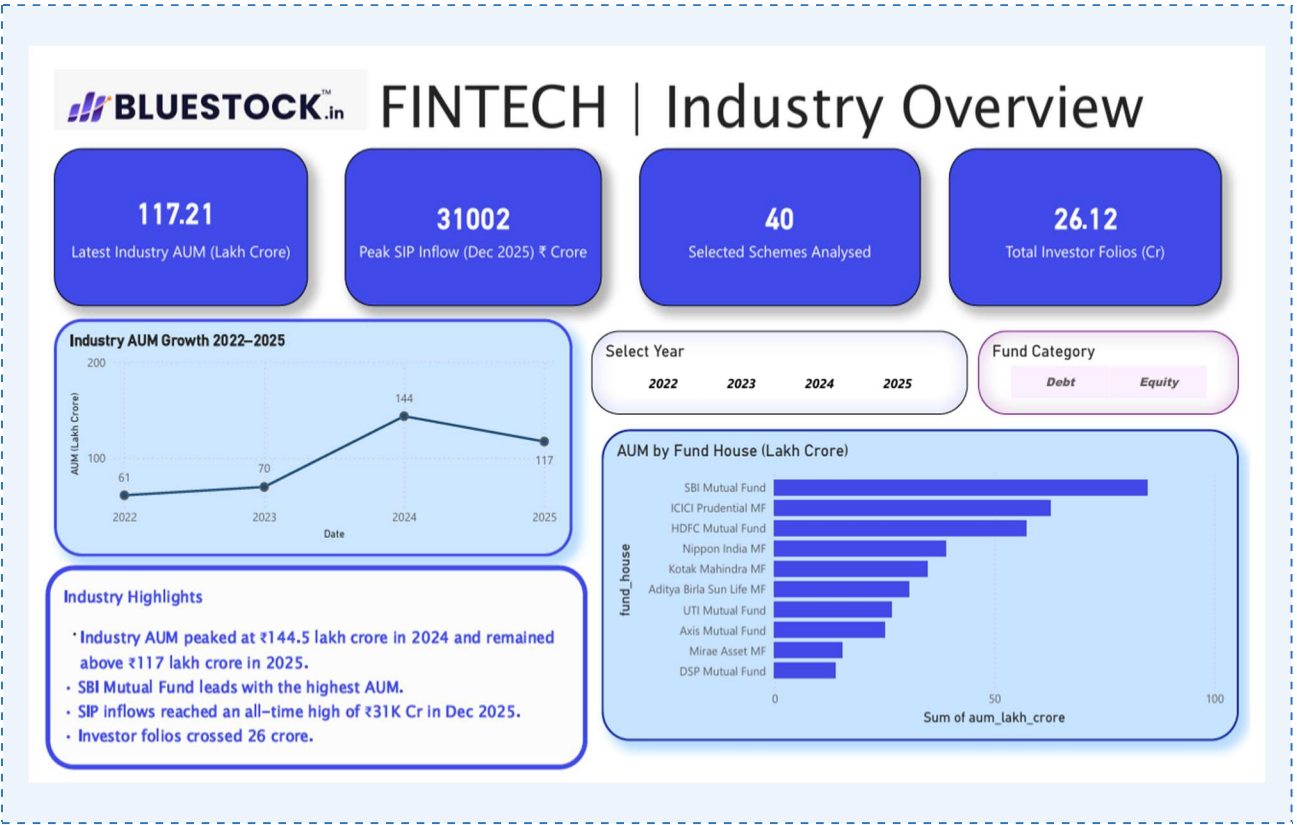
Run: `python scripts/recommender.py`

8. Power BI Dashboard

The dashboard consists of 4 interactive pages, each with a minimum of 2 slicers as required by the evaluation rubric. All data connects directly from cleaned CSVs in data/processed/.

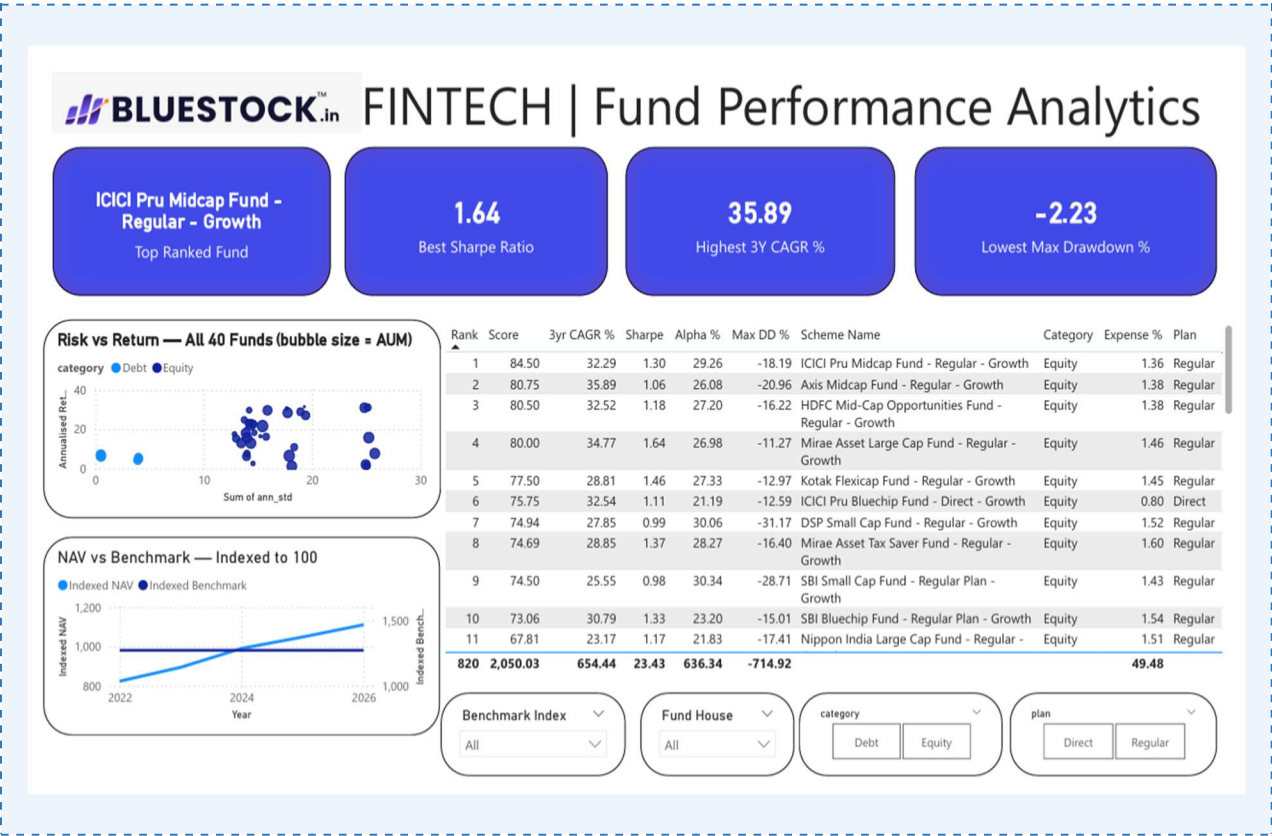
Page 1 — Industry Overview

Visual	Data Source	Slicers
4 KPI Cards (AUM, SIP Inflow, Folios, Schemes)	clean_aum_by_fund_house, clean_monthly_sip_inflows, clean_industry_folio_count	Year, Category
Line Chart: Industry AUM Growth 2022–2025	clean_aum_by_fund_house[date, aum_lakh_crore]	Year
Bar Chart: AUM by Fund House (Top 10)	clean_aum_by_fund_house[fund_house, aum_lakh_crore]	Year



Page 2 — Fund Performance

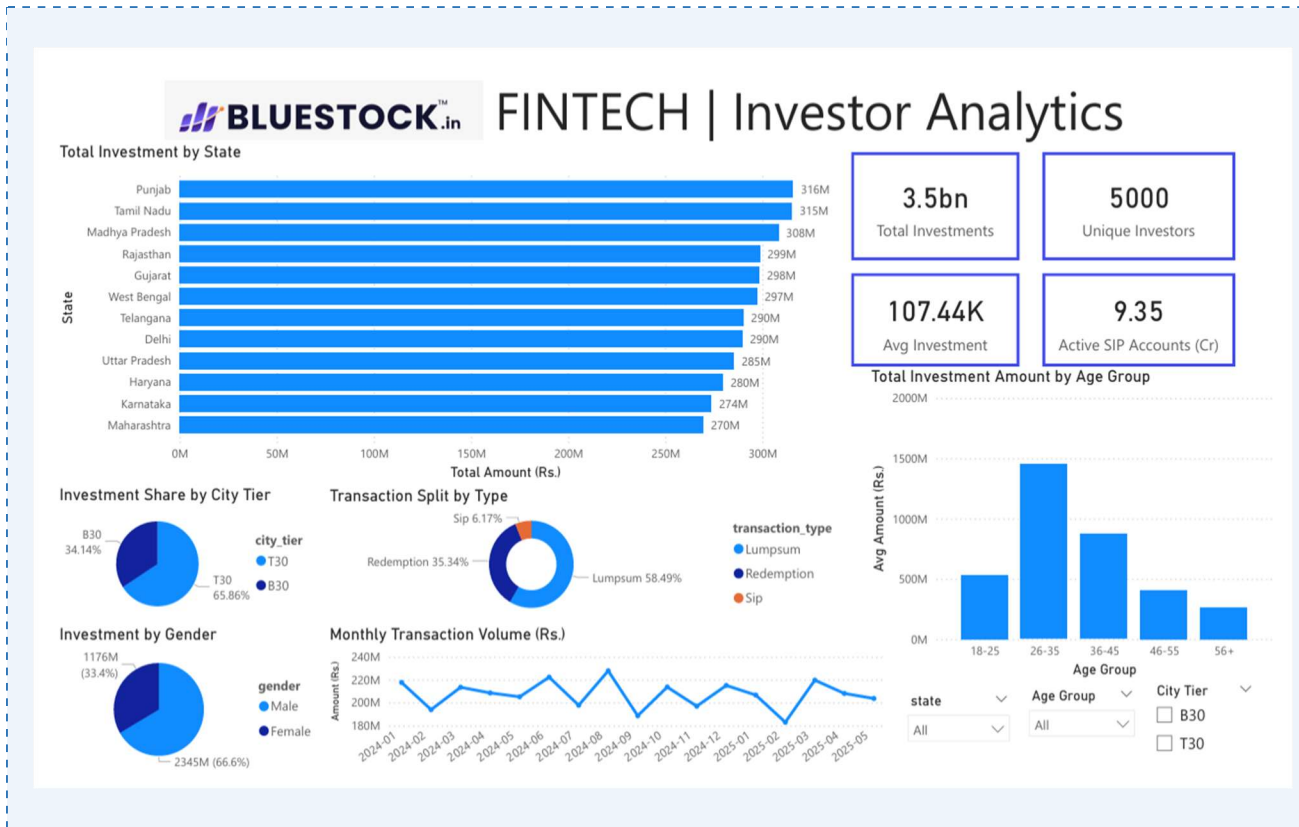
Visual	Data Source	Slicers
Scatter Plot: Risk vs Return (bubble = AUM)	sharpe_values[ann_std, ann_return], clean_scheme_performance[aum_crore]	Fund House, Category, Plan
Table: Sortable Fund Scorecard	fund_scorecard[all columns]	Fund House
Line Chart: NAV vs Benchmark (indexed to 100)	clean_nav_history, clean_benchmark_indices	Benchmark Index



Page 3 — Investor Analytics

Visual	Data Source	Slicers
Bar Chart: Transaction Amount by State	clean_investor_transactions[state, amount_inr]	State, Age Group, City Tier
Donut Chart: SIP vs Lumpsum vs Redemption	clean_investor_transactions[transaction_type]	State
Column Chart: Avg SIP Amount by Age Group	clean_investor_transactions[age_group, amount_inr]	City Tier

Visual	Data Source	Slicers
Line Chart: Monthly Transaction Volume	clean_investor_transactions[month, amount_inr]	State
Pie Chart: T30 vs B30 Gender Split	clean_investor_transactions[city_tier, gender]	Age Group



Page 4 — SIP & Market Trends

Visual	Data Source	Slicers
Dual Axis: SIP Inflow (bar) + Nifty 50 (line)	clean_monthly_sip_inflows, clean_benchmark_indices[NIFTY50]	Date Range
Matrix Heatmap: Category Inflows by Month	clean_category_inflows[category, month, net_inflow_crore]	Category
Bar Chart: Top 5 Categories by Net Inflow FY25	clean_category_inflows — Net Inflow FY25 measure	Date Range
4 KPI Cards: SIP growth metrics	clean_monthly_sip_inflows — DAX measures	—

149.05

New SIP Registrations 2025 (Lakh)

10.21

SIP Accounts YoY Growth %

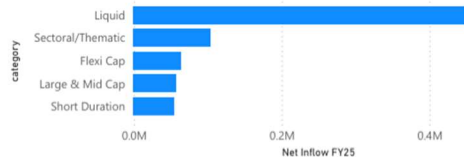
9.35

Active SIP Accounts (Crore)

336K

Total SIP Inflow 2025 (Crore)

Top 5 Categories by Net Inflow FY25



Category Inflows Heatmap (Rs. Crore)

category	2024	2025	Total
Liquid	104947	149475	451275
Sectoral/Thematic	25722	78107	103829
Flexi Cap	16438	47551	63989
Large & Mid Cap	14583	43169	57752
Short Duration	13671	41859	55530
Mid Cap	14196	41116	55312
Small Cap	12392	34204	46596
Hybrid	9157	29711	38868
Large Cap	6184	19449	25633
Value/Contra	4236	12744	16980
Gilt	2642	7753	10395
ELSS	1423	4627	6080
Total	225621	706618	932239

Fund Category

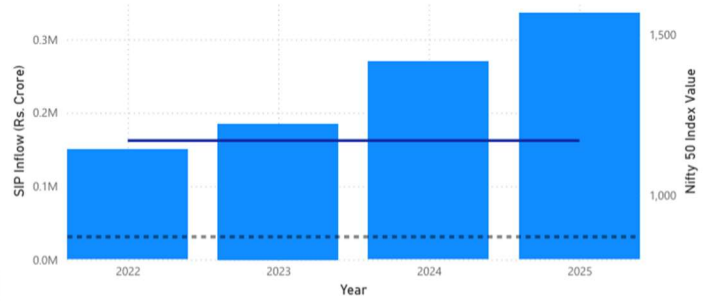
category
All

Select Date Range

month
01-01-2022 01-12-2025

SIP Inflow vs Nifty 50 (2022–2025)

Sum of sip_inflow_crore Nifty Return %



SIP Highlights

- New SIP registrations crossed 149 lakh in FY2025.
- Active SIP accounts reached 9.35 crore.
- Liquid funds attracted the highest inflows.
- Total SIP inflows exceeded ₹336K crore.

9. Key Recommendations

Based on the analytics built in this project, the following five recommendations are presented for Bluestock Fintech and its investors:

R1: Direct Plan Migration

Finding: Direct plans save 0.7–1.0% annually in expense ratio vs Regular plans. This compounds significantly over time.

Action: Educate investors to switch from Regular to Direct plans. On Rs. 1 lakh invested at 12% return over 10 years, saving 0.8% expense ratio = Rs. 28,000 additional corpus.

R2: SIP Continuity Intervention

Finding: A significant proportion of investors with 6+ SIP transactions show avg gap > 35 days, indicating irregular SIP behaviour.

Action: Send automated reminders to at-risk investors 30 days after their last SIP. Improving continuity increases rupee cost averaging benefit and long-term returns.

R3: Small Cap Risk Warning

Finding: Small Cap funds deliver highest returns (CAGR 27–35%) but worst drawdowns — SBI Small Cap Direct suffered -52.57% max drawdown.

Action: Implement mandatory risk profiling before recommending Small Cap funds. Map only 'Very High' risk appetite investors to this category — aligns with SEBI categorisation.

R4: B30 City Expansion

Finding: B30 cities account for only 33.7% of transactions despite significant investor population in Tier 2/3 cities.

Action: Launch targeted digital SIP campaigns in B30 cities. B30 investors show consistent avg SIP amounts. AMFI data shows B30 penetration is a key growth vector industry-wide.

R5: Portfolio Concentration Risk Awareness

Finding: Axis Bluechip has HHI of 2,968 (highly concentrated in 7 sectors). UTI Mid Cap is most diversified (HHI 1,240 across 10 sectors).

Action: Recommend portfolio diversification by combining high-HHI Large Cap funds with low-HHI Flexi Cap or Mid Cap funds. Better sector diversification reduces portfolio volatility.

10. Limitations & Future Work

Current Limitations

L1: Dataset Scope

Only 40 of 1,908 industry schemes analysed. Covers top schemes from 10 AMCs — representative but not exhaustive.

L2: Simulated Transaction Data

08_investor_transactions.csv is synthetically generated. Real investor data is confidential under SEBI regulations. Uses real geographic and demographic distributions.

L3: 5-Year CAGR Unavailable

Data starts January 2022 — only 4.4 years available. 5-year CAGR computed as proxy using full available range.

L4: Benchmark Correlation

NAV data is simulated forward from real anchor values. Low R-squared in Alpha/Beta OLS regression is expected. Pre-computed values in scheme_performance.csv more reliable for Alpha/Beta.

L5: No Real-Time Refresh

Dashboard uses static CSVs. Live refresh requires Power BI Service or PostgreSQL with scheduled ETL pipeline.

Future Work

F1: Full Industry Coverage

Expand from 40 to all 1,908 AMFI schemes. Requires scheduled API fetch for all scheme codes.

F2: Monte Carlo Simulation

Project NAV growth over 5 years with uncertainty bands (Bonus B3). Uses historical volatility to generate 1,000+ simulated paths.

F3: Markowitz Efficient Frontier

Portfolio optimisation for 5 selected funds (Bonus B4). Compute optimal weights for maximum Sharpe ratio portfolio.

F4: Streamlit Web App

Build Streamlit alternative to Power BI (Bonus B2). Enables online deployment without Power BI license.

F5: PostgreSQL Production

Migrate from SQLite to PostgreSQL for production. Enables concurrent users, larger datasets, and REST API access.

F6: HTML Email Reports

Automated weekly performance summary emails (Bonus B5). Triggered from scheduled ETL pipeline output.

11. Technical Stack & File Structure

11.1 Technology Stack

Category	Tool / Library	Version	Purpose
Language	Python	3.10+	All data processing, ETL, analytics, scripting
Data Manipulation	Pandas	2.0+	DataFrames, data cleaning, aggregation, merging
Numerical	NumPy	1.24+	Risk metrics, statistical computations, arrays
Statistics	SciPy	1.10+	OLS regression for Alpha/Beta (linregress)
Visualisation	Matplotlib	3.7+	Static publication-quality charts for reports
Visualisation	Seaborn	0.12+	Heatmaps, correlation matrices, distributions
Visualisation	Plotly	5.x	Interactive charts in Jupyter notebooks
Database	SQLite3	Built-in	Local relational database (no server needed)
ORM	SQLAlchemy	2.0	Python-to-SQLite interface (df.to_sql)
Notebooks	Jupyter Lab	4.x	Interactive EDA and analytics environment
Dashboard	Power BI	Latest	4-page interactive business intelligence dashboard
Version Control	Git + GitHub	Latest	Code versioning and project submission
API	mfapi.in	v1	Live NAV data — no authentication required
HTTP Client	Requests	2.30+	Python HTTP library for mfapi.in API calls
IDE	VS Code	Latest	Development environment

11.2 Project File Structure

File / Folder	Purpose	Day
scripts/data_ingestion.py	Loads all 10 CSVs, prints shape/dtypes/head	Day 1
scripts/live_nav_fetch.py	Fetches live NAV from mfapi.in API for 6 funds	Day 1
scripts/scheduled_nav_fetch.py	Bonus B1 — auto-fetch weekdays 8PM with logging	Day 1
scripts/etl_pipeline.py	Runs schema.sql, loads all cleaned CSVs to SQLite	Day 2
scripts/compute_metrics.py	Computes all 7 metrics from raw NAV history	Day 4
scripts/recommender.py	Fund recommendation engine by risk appetite	Day 6
notebooks/01_data_ingestion.ipynb	Interactive Day 1 notebook — outputs, EDA cells	Day 1
notebooks/02_data_cleaning.ipynb	12-cell cleaning notebook — all 10 CSVs	Day 2
notebooks/sql_queries.ipynb	10 analytical SQL queries with DataFrame results	Day 2
notebooks/03_eda_analysis.ipynb	15+ charts — NAV, AUM, SIP, demographics, sectors	Day 3
notebooks/04_performance_analytics.ipynb	7 metrics computed + scorecard + benchmark chart	Day 4
notebooks/05_advanced_analytics.ipynb	VaR, cohort, SIP continuity, HHI, recommender	Day 6
sql/schema.sql	CREATE TABLE statements for all 10 tables	Day 2
sql/queries.sql	10 analytical SQL queries	Day 2
data/db/bluestock_mf.db	SQLite database — 87,533 rows across 10 tables	Day 2
data/processed/fund_scorecard.csv	40 funds ranked by composite score (our pipeline)	Day 4
dashboard/bluestock_mf.pbix	Power BI dashboard — 4 pages, 10+ visuals	Day 5
data_dictionary.md	Business definitions for every column in every table	Day 2
README.md	Setup instructions, file descriptions, how to run	Day 7

12. Conclusion

This capstone project successfully demonstrates the complete data engineering and analytics lifecycle — from raw CSV ingestion to an interactive executive dashboard — applied to India's rapidly growing mutual fund industry.

Project Achievement Summary

87,533 rows loaded into a 10-table SQLite database. 7 performance metrics computed from raw NAV history using our own pipeline. 15+ EDA charts documenting 4.4 years of market data. A 4-page Power BI dashboard enabling self-service analysis. Fund recommendation engine covering all 5 SEBI risk categories.

7/7

Deliverables Completed

8/8

Objectives Met

B1 Done

Bonus Features

Modular

Code Quality

The platform mirrors real-world fintech data engineering practices used at companies like Zerodha, Groww, and Paytm Money — demonstrating skills in Python data engineering, SQL database design, statistical computing, and business intelligence visualisation.

Bluestock Fintech Pvt. Ltd. | Capstone Project | Mutual Fund Analytics Platform

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